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1. Define computer, who is the father of computer?

Computer is a electronic device that is used for information processing it accepts data and instruction stores it in a memory process and gives the result to the user.

Father of computer is Charles babbage

b) List the characteristics & capabilities of computer?

Characteristics	Capability
High processing speed	Huge data storage
Accuracy	Input and output processing
Reliability	
versatality	
Diligence	

c) Define printer and list its types.

A printer prints the output information from the computer on to a paper printers are generaly used to print textual information, but know a day printers are also print graphical information.

Types :

- Impact printer
- Non-impact printer

d) Write any 2 difference blw main memory & secondary memory

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Main memory
- Main memory is a temporary
- Hold data & instruc -tions till the power supply lasts

secondary memor
- It is a permanent memor
- Hold data & instruc -tions as long as the user can

e] write the structure of Basic C-program

```
# Include <stdio.h>
int main ()
{
printf ("Hello word ");
return 0;
}
```

4] Define C-program who is the father of C is a high-level-language it is both general purpose and a specific purpose programming language
Father of C-program is - Dennis Ritchie

Q II

5] list & explain types of computer
① Analog computer
② Digital computer
③ Hybrid computer.

① Analog computers :- Analog computer is a work upon continous data analog computers operate by measuring rather the counting. the devices that measures such quantities are analog computer.

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(2) Digital computer :- The digital computer works upon discountinuous data. A digital computer operates on digital data such as number.

(3) Hybrid Computers :- Hybrid computer are the combination of both analog and digital computers. They accept both the analog & digital data for processing.

(3) Explain Generations of computers?

- First Generation Computers :- These were developed during 1946-58, and were build with vaccum tubes. their speed was 10^{-3} sec. They had limited primary memory and used magnetic drums as secondary storage devices
  Ex :- ENIAC, UNIVAC-I

- second Generation :- These were developed during 1957-64 and were build with diodes and transistors their speed was 10^{-6} sec. They had more primary memory and provided higher reliability
  Ex :- IBM.

- 3rd Generation :- 1965-1970 developed (IC) integrated circuit is being used to develop device 10-100 transister. They used semiconductor memory.
  Ex :- IBM system 360.

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- 4th Generation :- 1970 to 1990 they use (LSI) large scale integration to develop this computer 100-1000 number of transistors are used to very large scale integration speed is 10^-9 to 10^-12 Ex:- IBM 3033 HD-3000

5th Generation :- (SLSI) super large scale integration device used this computer, millions of transistor is used to develop IA this computer Ex:- cray, PARAM.

4] explain exchange values.

```
#include <stdio.h>
int main()
{
int a, b;
scanf("%d", &a);
scanf("%d", &b);
printf("before swopping %d and %d\n"
        , a, b);
a = a+b;
b = a-b;
a = a-b;
printf("after swopping %d and %d", a, b);
    return 0;
}
```

5] write a c-program to calculate addition of 2 numbers

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```
# include <stdio.h>
int main ()
{
int a = 10;
int b = 20;
int sum;
sum = a+b;
printf ("sum = %d", sum);
return 0;
}
```

6] Discuss history & evolution of computer.

- Abacus :- This is the first recorded computer, whose existence dates back to 2500 B.C

- Napier's Bones :- In 1614, a scottish mathematician napier introduced the concept of logarithms. He used a set of bones to perform multiplication. He used 11 bones having four faces each.

- slide rule :- In 1630, an english mathematician, William oughtred, developed the slide rule. It was designed on the basis of logarithms. It consists of two scales one of which slides over the other

- pascaline :- In 1642, a french mathematician, Blaise pascal invented this machine. He developed it for his father's business calculations.

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Jacquard's loom :- In 1802, a French textile manufacturer, joseph marie Jacquard, invented a machine which was used to automatically control the weaving loom. jacquard employed punched cards to control the patterns of woven cloth

III

8] Explain von-Neumann Concept of computer with block diagram.

Basic structure of computer.

- Accepting data and information from the user An input device is used to accept data and instructions
- storing data and information in its memory the memory devices are used to store the data and instructions in the computer.
- processing the data and information. The central processing unit (CPU) process the data
- Displaying the processed data, finally the monitor (visual display unit) or computer screen will display the desired results.

Input device :-

- This is used to feed data and instructions into the computer. it is connected to the system unit.
- other input devices are mouse, light pen, joystick, touch panel, microphone, disks, digital cameras & voice input

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System unit

- This is responsible for storing and processing of data and instructions.
- The CPU is the main unit in a computer system, which performs all arithmetic and logical operations.

Functions of the control unit

- Fetching data and instructions from the memory
- Interpreting the instructions
- Controlling the transfer of data and instruction to and from the main memory
- Controlling the input and output devices
- The overall supervision of a computer system

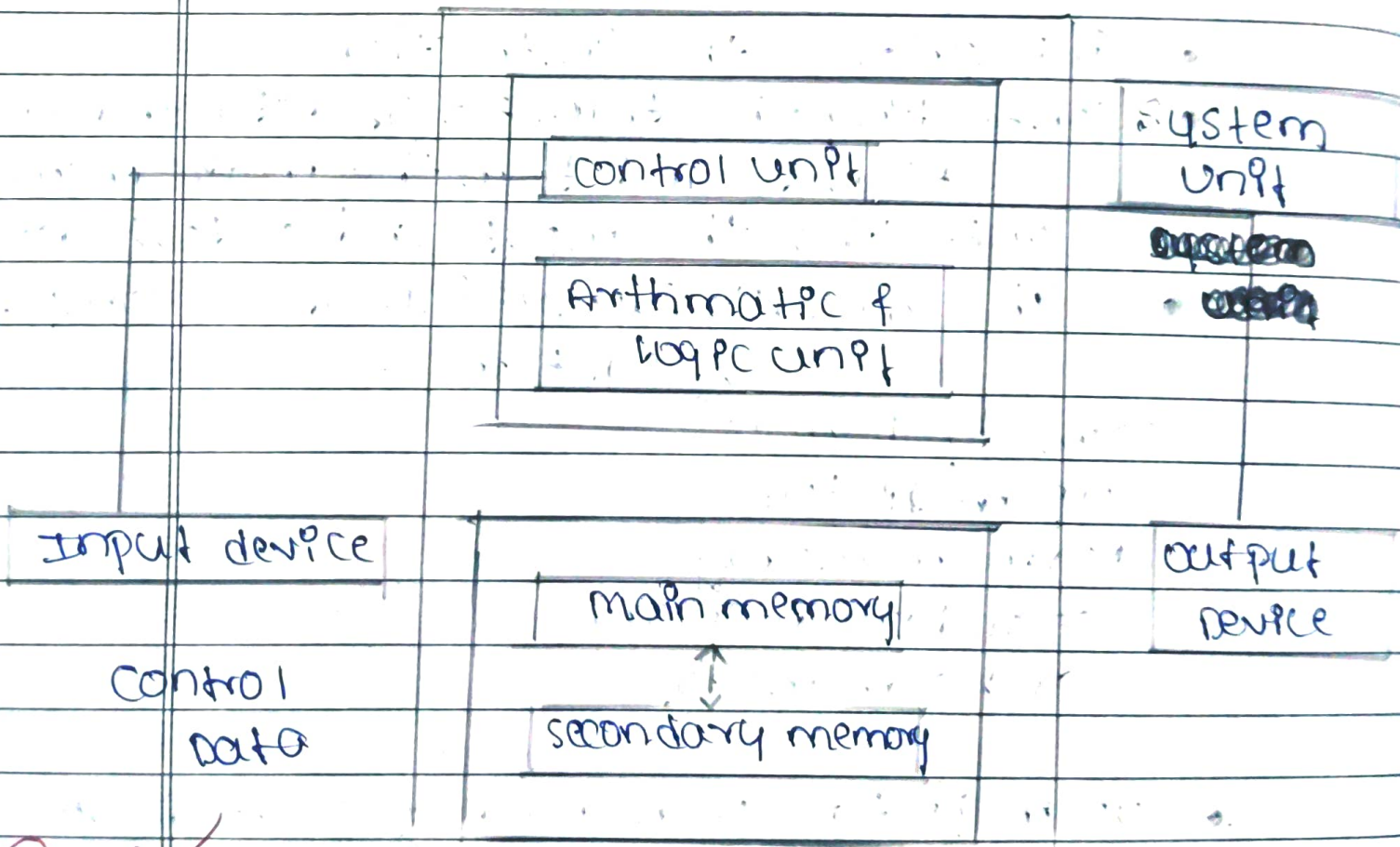
Memory devices

CPU registers
Main memory
Secondary memory
Cache memory.

- CPU registers :- It hold a limited amount of memory during execution. They are inside the CPU.
- main memory :- is the primary memory which holds data & instruction supplied by the user it is temporary memory
- secondary memory :- is a permanent memory which holds data and instructions as long as the user can do.
- cache memory :- is a high-speed memory and comes in blw the CPU and the main memory

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output device.
once the data and instructions are processed the user can choose to display such results on the output devices.
- The monitor is the most commonly used output device there are several other devices which are used as the output devices
Ex:- printers, plotters, plasma panel display (PPD), LCD-s, speakers & disks.



24/09/26